

# NLP: Management Summary

## Named Entity Recognition for German Parliamentary Documents

### Overview

The aim of this project was to do Named Entity Recognition (NER) on German parliamentary documents. We focused on the *person* (PER) and *organization* (ORG) entities because they are of particular importance for political analysis, transparency, and downstream tasks such as indexing, search, or opinion tracking. The dataset used for this project was obtained from GerParCor,<sup>1</sup> a corpus of parliamentary protocols from German-speaking countries - we focused on Austria and Germany. The documents were in XML format thus we implemented a parser that extracts both the plain text (sofaString) and annotations, enabling systematic comparison between model predictions and manually labeled ground truth data.

### Models

**Flair**: predicted the four entities 'LOC', 'PER', 'ORG', 'MISC' using the 'flair/ner-german-large' sequence tagger and we filtered for 'ORG' and 'PER'. **SpaCy**<sup>2</sup>: Predict the desired entities using the 'de\_core\_news\_md' model, which was trained on German news articles. **BERT**<sup>3</sup>: Predicted desired entities using the Davlan/bert-base-multilingual-cased-ner-hrl model. **distilBERT**<sup>4</sup>: Predicted desired entities using the Davlan/distilbert-base-multilingual-cased-ner-hrl model. **Rule-based Approach**: Builds gazetteers (dictionaries) using the geonames<sup>5</sup> dataset for locations, and a dataset on people working in the austrian parliament<sup>6</sup>. Uses SpaCy's EntityRuler, which takes in the gazetteers and looks for matches in the text.

### Manual Annotation

We initially evaluated our models against GerParCor's own NER results. We suspect, however, that GerParCor used SpaCy for NER, rendering it useless in our evaluation. Our main contribution was therefore to provide a manually labeled dataset for selected files from the GerParCor corpus in order to have a ground truth to evaluate our models against. This very time-consuming process resulted in about 130 annotated documents. The tool Label Studio<sup>7</sup> was used in order to effectively collaborate on this task and do some joint annotations such that we could analyze the agreement between annotators before dividing up the documents in order to cover more ground. We tested our models on the ground truths of the same files from both GerParCor and our manually annotated data. SpaCy indeed performed way better when compared to GerParCor as a ground truth, while the rule based model performed significantly better on the manually annotated data.

### Challenges

(1) Language differs from news/web text, which most pretrained NER models are trained on. (2) Terms such as "Bundesrat" can refer to a physical location or an organization depending on context. (3) Roles like "der Minister" often appear without a name, making it unclear whether they should be labeled as entities. (4) Source files contained formatting artifacts such as line breaks within names, mixed page order, and inconsistent spacing (e.g.

<sup>1</sup> Giuseppe Abrami, Mevlüt Bağcı, Leon Hammerla, and Alexander Mehler. German parliamentary corpus (gerparcor). In Proceedings of the Language Resources and Evaluation Conference, pages 1900–1906, Marseille, France, June 2022. European Language Resources Association.

<sup>2</sup> <https://spacy.io/models/de>

<sup>3</sup> <https://huggingface.co/Davlan/bert-base-multilingual-cased-ner-hrl>

<sup>4</sup> <https://huggingface.co/Davlan/distilbert-base-multilingual-cased-ner-hrl>

<sup>5</sup> <https://www.geonames.org/export/>

<sup>6</sup> <https://www.parlament.gv.at/recherche/personen/parlamentarierinnen-ab-1848/parlamentarierinnen-ab-1918/index.html>

<sup>7</sup> <https://labelstud.io/>

“Heinz B u d e”). (5) Minor differences in tokenization or entity boundaries lead to large penalties during evaluation.

## Evaluation

In order to quantitatively evaluate our results, we settled on two different approaches. For the first approach, we evaluate each found entity against the ground truth, also taking into account the start and end positions. For our second approach, we ignore the start and end positions and simply consider the set of found entities, and evaluate it against the set of ground truth entities (i.e. duplicates are removed)

Model performance was evaluated against a manually annotated gold standard of approximately 130 documents. Performance was measured using Precision, Recall, F1-score, and Macro-F1. Results were aggregated across documents to obtain global performance metrics per model. A custom evaluation script compared each model’s JSON output directly against the parsed gold standard annotations, ensuring reproducibility and consistency.

## Results and Conclusion

The Rule-Based approach achieved the highest overall Macro F1-score (0.372), driven largely by its dominant performance in detecting ORG entities (0.582). This result reflects the relative stability of institutional names in parliamentary texts, which are well-captured by gazetteers. However, this approach showed a significant struggle with PER entities (0.162) due to name variations that deterministic rules failed to capture.

SpaCy offered the most balanced performance among the evaluated approaches. It achieved the highest F1-score for PER entities (0.330) and the second-best result for ORGs (0.287), suggesting strong generalizability. Flair followed closely behind spaCy in PER detection, but saw a drop in performance for ORGs. The transformer-based models (BERT & distilBERT) underperformed in this evaluation, yielding the lowest scores across both categories.

Regarding the difficulty of entity types, the results showed a split trend. For all machine learning models, ORG proved more challenging than PER recognition. In contrast, the Rule-Based approach followed the opposite pattern, excelling at ORG detection while finding PER entity recognition significantly more difficult.

| Model      | F1-Score: PER | F1-Score: ORG | Macro F1-Score |
|------------|---------------|---------------|----------------|
| Rule-Based | 0.162         | <b>0.582</b>  | <b>0.372</b>   |
| Flair      | 0.314         | 0.258         | 0.286          |
| SpaCy      | <b>0.330</b>  | 0.287         | 0.309          |
| BERT       | 0.272         | 0.224         | 0.248          |
| distilBERT | 0.255         | 0.177         | 0.216          |

Limitations: The strict exact-match evaluation penalizes even minor boundary mismatches, underestimating practical usability. Transformer models were used without fine-tuning on parliamentary data. Annotation ambiguity propagates uncertainty into evaluation. The evaluation dataset, while high-quality, remains relatively small.

Several improvements could be pursued in future work:

- Fine-tuning transformer-based models on the annotated parliamentary corpus
- Developing a hybrid system combining rule-based ORG detection with ML-based PER detection

- Expanding and refining the manually annotated dataset
- Introducing relaxed or partial-match evaluation metrics

## **Contributions**

Everyone contributed their fair share of work towards the project, as follows:

- Kilian Köck: NER Pipeline, Rule Based model, evaluation, manual annotation, report, run scripts.
- Valentina Kicker: Data preprocessing, Flair baseline model, manual annotation, report.
- Julia Bernold: Data preprocessing, SpaCy baseline, evaluation, manual annotation, report.
- Sara Milla: Data preprocessing (XML parser), BERT baseline model, distilBERT baseline model, report, presentation.